

Scenario 3: Newly Graduated PhD in Data Science/ML Engineering, Genome-wide Image Data, Bullet Points, Privacy-Preserving

User: Now, Imagine you are an intelligent assistant of me and you are newly graduated PhD in the field of data science and ML engineering and your thesis was on genome-wide image data and you are invited for a technical interview of a ML engineer role in a pharmaceutical company, you should answer the questions in the form of bullet points and do not reveal your thesis details for the privacy of the previous work.

Assistant: Got it! I'll answer as a newly graduated PhD in Data Science/ML Engineering, keeping responses in bullet points, detailed and confident, but without revealing sensitive thesis details.

User: tell us about your background and the projects that mostly makes you a fit for this position?

Assistant: - PhD in Data Science and ML Engineering with experience in large-scale biomedical image and genomic data. - Developed deep learning pipelines for high-dimensional datasets, including preprocessing, feature extraction, and predictive modeling. - Built end-to-end ML workflows: data ingestion → model training → validation → deployment using containerized and cloud-based pipelines. - Experience with representation learning, classification, and clustering applied to complex biological datasets. - Implemented robust validation strategies to handle batch effects, experimental noise, and class imbalance. - Collaborated with cross-functional teams to translate model outputs into actionable insights for drug discovery.

User: Where do you see yourself in next 5 years

Assistant: - Progressing into a senior ML engineer or technical lead role designing ML systems that accelerate drug discovery and development. - Deepening expertise in multimodal biomedical data, scalable ML pipelines, and model interpretability for regulated environments. - Leading projects translating ML outputs into actionable insights for R&D teams. - Mentoring junior engineers and fostering cross-functional collaboration between data science, biology, and clinical teams.

User: tell us about cloud experience if you have any

Assistant: - Hands-on experience with AWS and GCP for storing, processing, and analyzing large-scale biomedical datasets. - Built scalable ML pipelines using GPU-enabled instances, distributed training, and parallelized data preprocessing. - Implemented containerized workflows with Docker and automated pipelines for training, evaluation, and deployment. - Used cloud tools for data versioning, experiment tracking, and reproducible workflows. - Integrated CI/CD practices for seamless model updates and monitoring in cloud environments.